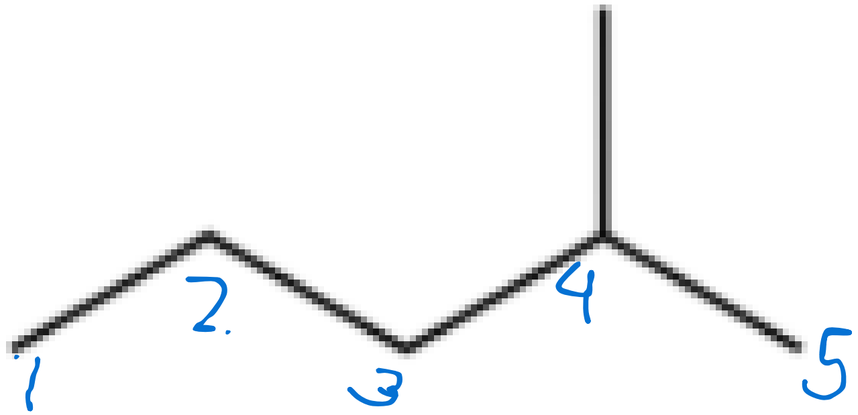
CARD A01

Foundation | Formula: C6H14

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 5 carbons → pentane

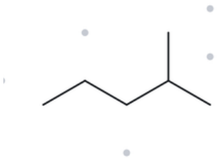
Numbering: L-to-R gives locant 2; R-to-L gives 4. Lower set {2} wins.

Substituents: Methyl at C2 → 2-methyl

3. MY NAME (WRITTEN FIRST)

2-methylpentane

4. THE ALKANE CHECK



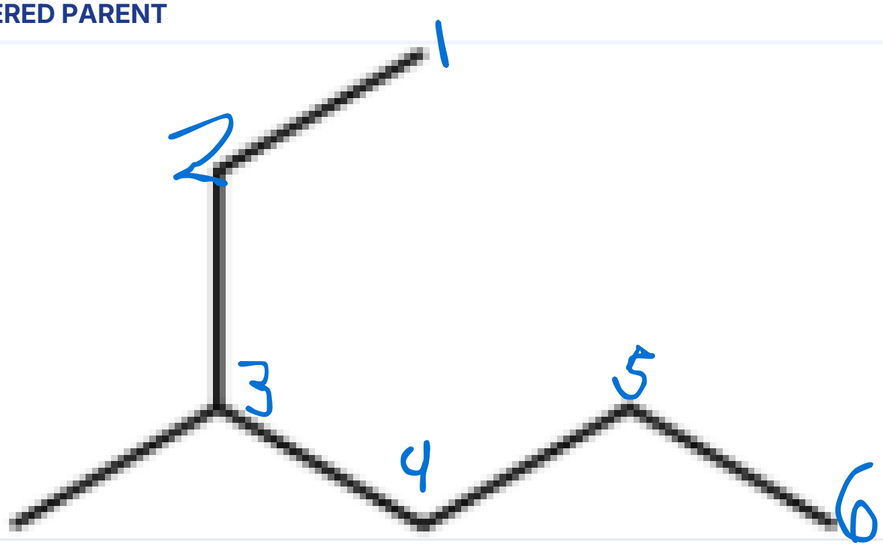
✓ Match

Returned: 2-methylpentane

2-methylpentane

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Bent chain = 6 carbons → hexane

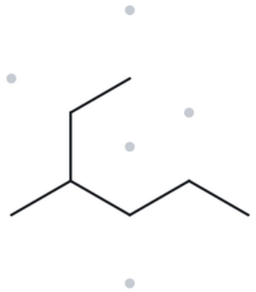
Numbering: Bottom-left to right gives locant 3 (wins over 4).

Substituents: Methyl at C3 → 3-methyl

3. MY NAME (WRITTEN FIRST)

3-methylhexane

4. THE ALKANE CHECK



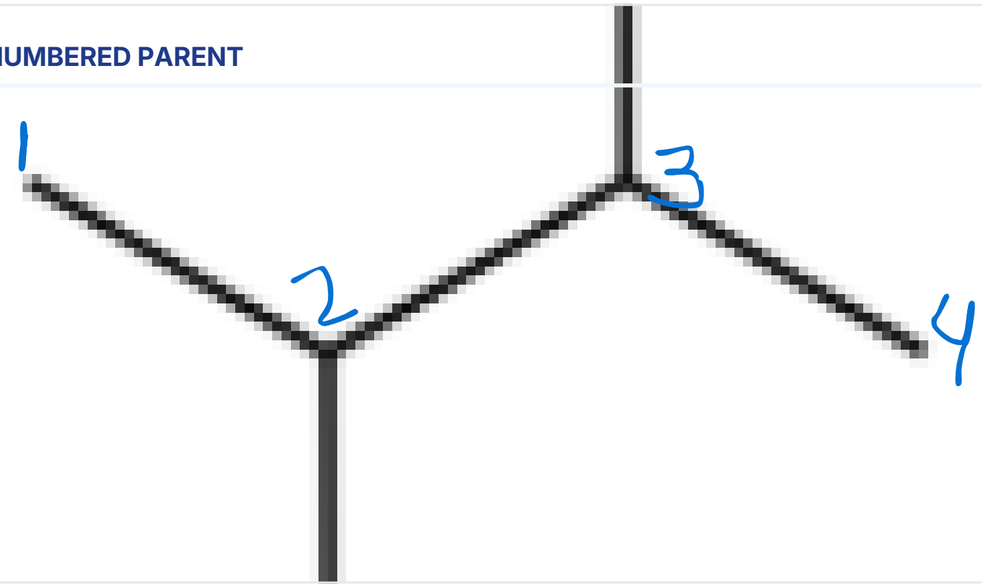
✓ **Match**

Returned: 3-methylhexane

3-methylhexane

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 4 carbons → butane

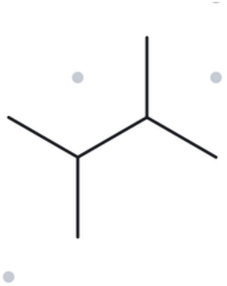
Numbering: Symmetrical set {2,3} from either end.

Substituents: Methyls at C2, C3 → 2,3-dimethyl

3. MY NAME (WRITTEN FIRST)

2,3-dimethylbutane

4. THE ALKANE CHECK



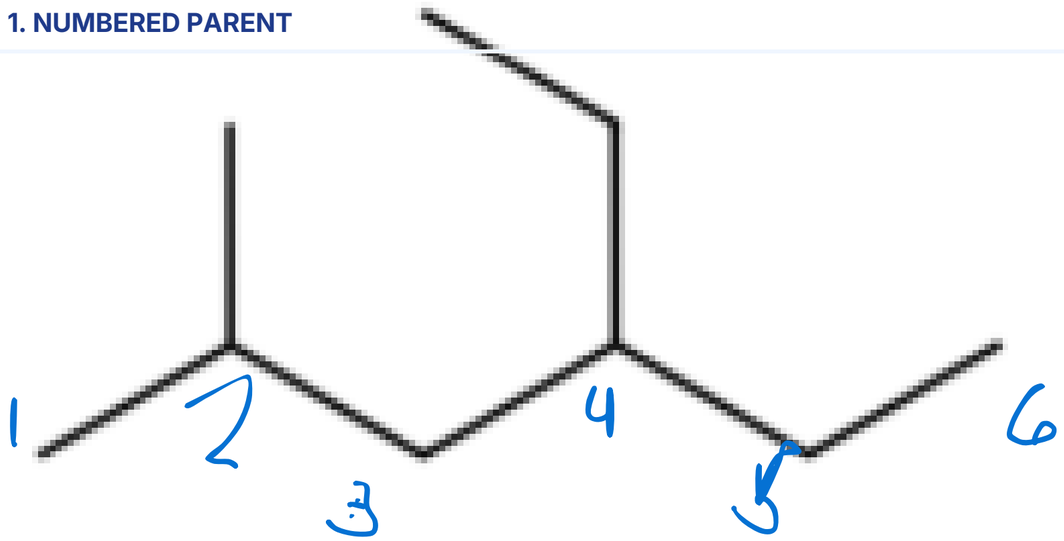
✓ **Match**

Returned: 2,3-dimethylbutane

2,3-dimethylbutane

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 7 carbons → heptane

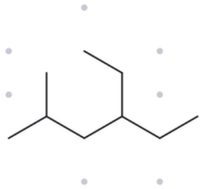
Numbering: L-to-R gives {3,4}; R-to-L gives {4,5}. {3,4} wins.

Substituents: Ethyl (C4), Methyl (C2) → ethyl cited first

3. MY NAME (WRITTEN FIRST)

4-ethyl-2-methylhexane

4. THE ALKANE CHECK



4-ethyl-2-methylhexane

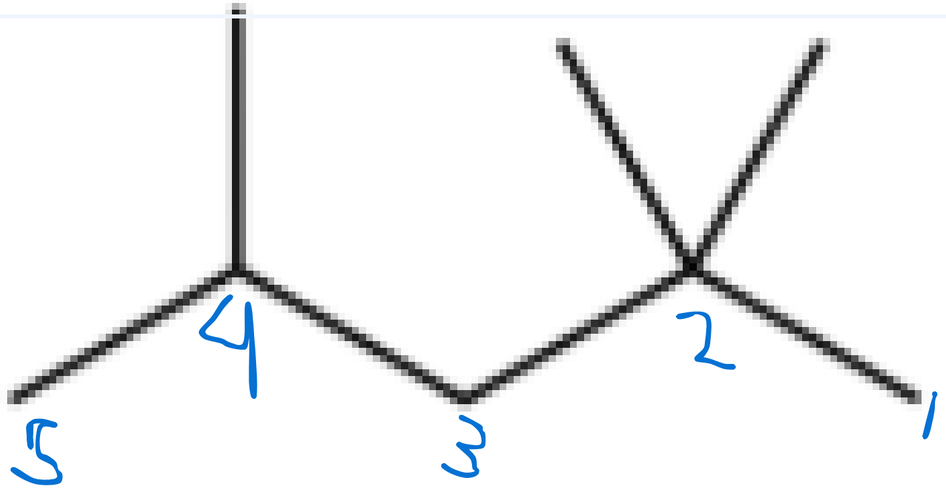
✓ Match

Returned: 4-ethyl-2-methylhexane

CARD A05

Developing | Formula: C8H18

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 5 carbons → pentane

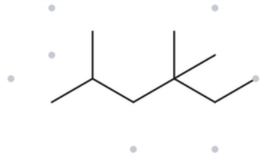
Numbering: {2,2,4} vs {2,4,4}. {2,2,4} wins at 2nd locant (2 < 4).

Substituents: Methyls at C2, C2, C4 → 2,2,4-trimethyl

3. MY NAME (WRITTEN FIRST)

2,2,4-trimethylpentane

4. THE ALKANE CHECK



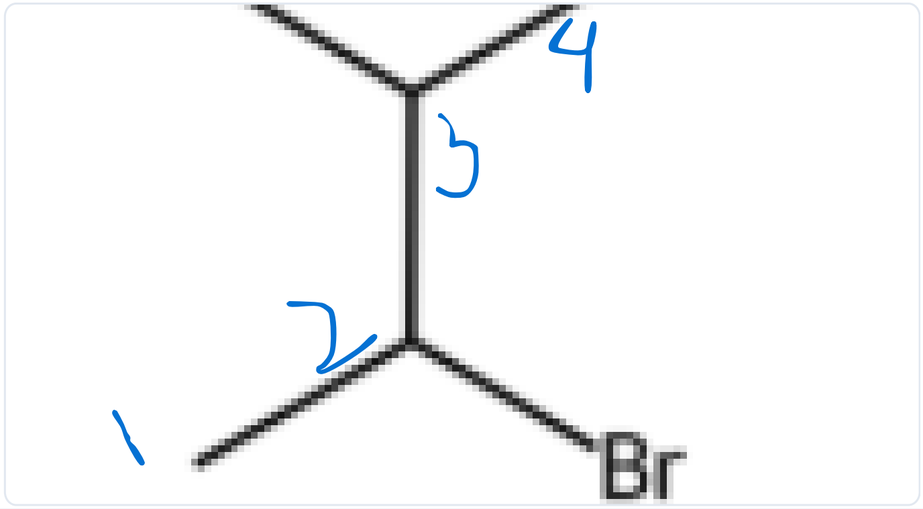
2,4,4-trimethylhexane

✓ **Match**

Returned: 2,2,4-trimethylpentane

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 5 carbons → pentane

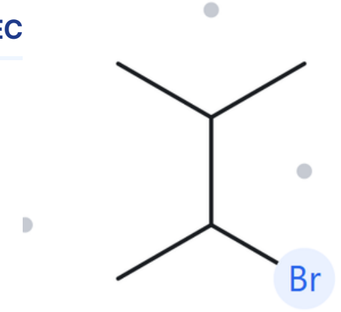
Numbering: Locant tie {2,3}. Alphabetical tie-break: bromo gets C2.

Substituents: 2-bromo, 3-methyl

3. MY NAME (WRITTEN FIRST)

2-bromo-3-methylbutane

4. THE ALKANE CHEC

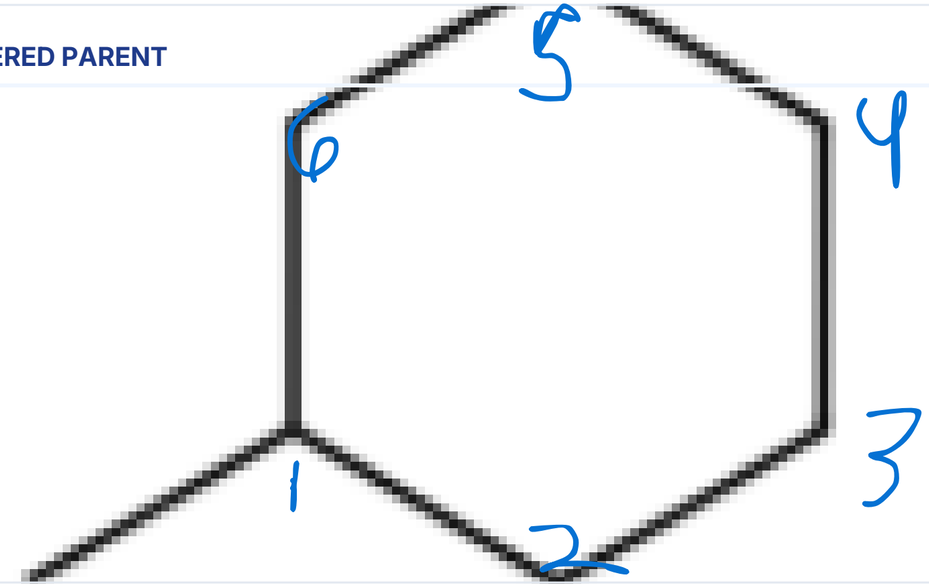


2-bromo-3-methylbutane

✓ Match

Returned: 2-bromo-3-methylbutane

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: 6-membered ring → cyclohexane

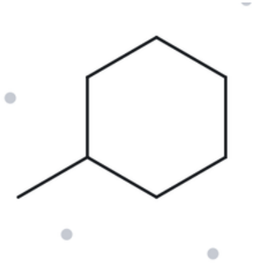
Numbering: Monosubstituted ring; locant 1 is omitted.

Substituents: Methyl group

3. MY NAME (WRITTEN FIRST)

methylcyclohexane

4. THE ALKANE CHECK



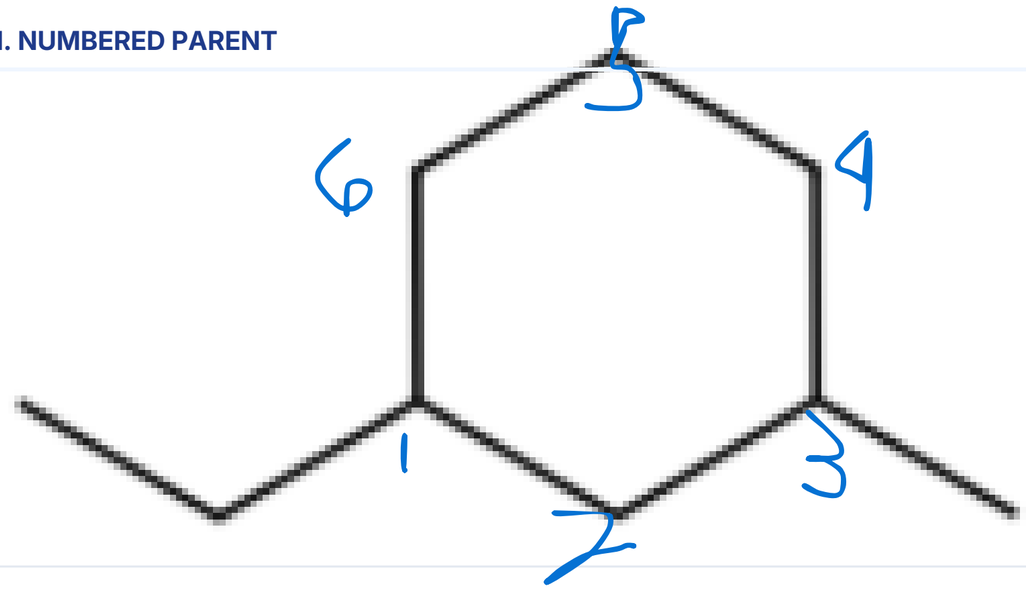
methylcyclohexane

✓ Match

Returned: methylcyclohexane

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: 6-membered ring → cyclohexane

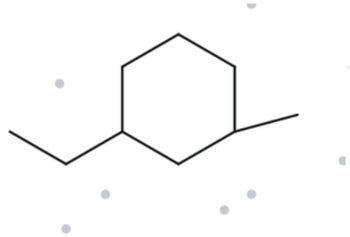
Numbering: 1,3-tie broken alphabetically: ethyl gets C1.

Substituents: 1-ethyl, 3-methyl

3. MY NAME (WRITTEN FIRST)

1-ethyl-3-methylcyclohexane

4. THE ALKANE CHECK



✓ Match

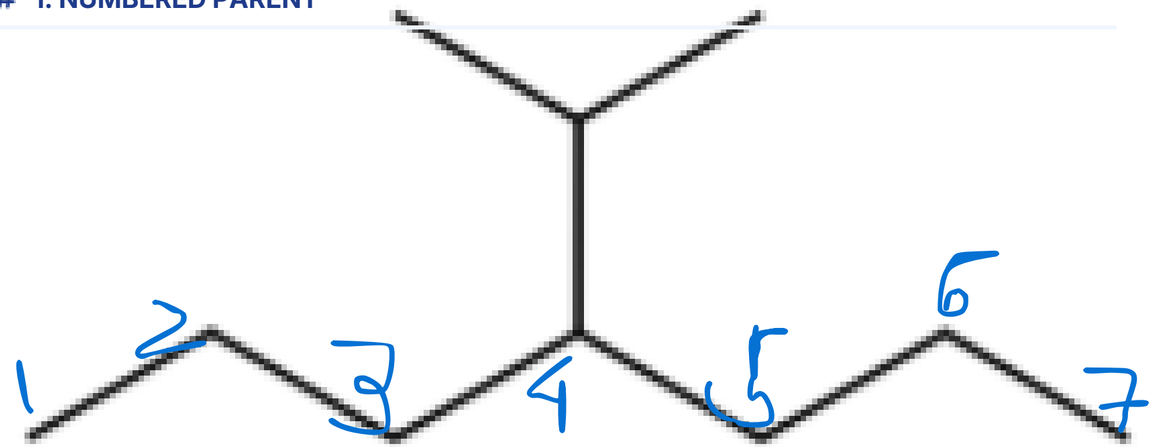
1-ethyl-3-methylcyclohexane

Returned: 1-ethyl-3-methylcyclohexane

CARD A09

Challenge | Formula: C10H22

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: Longest chain = 7 carbons → heptane

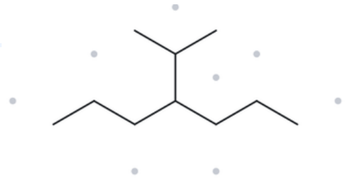
Numbering: Locant 4 from either direction.

Substituents: Isopropyl group at C4

3. MY NAME (WRITTEN FIRST)

4-isopropylheptane

4. THE ALKANE CHECK



2-methyl-3-propylhexane

Match

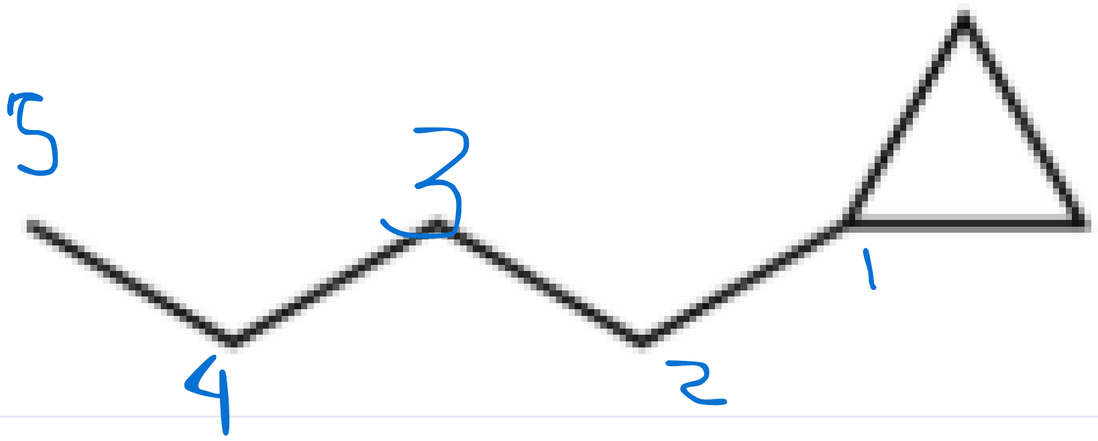
Returned: 4-(1-methylethyl)heptane (Equivalent form)

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A10

Challenge | Formula: C8H16

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: 3-membered ring is parent → cyclopropane

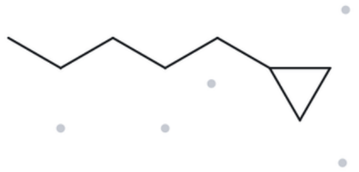
Numbering: Monosubstituted ring (locant unnecessary).

Substituents: 5-carbon chain → pentyl

3. MY NAME (WRITTEN FIRST)

pentylcyclopropane

4. THE ALKANE CHECK



✓ Match

Returned: pentylcyclopropane

pentylcyclopropane

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD B01 NAME → STRUCTURE

Foundation | Given: 2-methylpropane

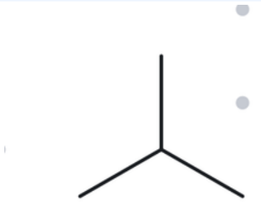
1. NAME DECODED

Parent: propane (3-carbon chain)

Substituents: methyl at C2

Expected Formula: C4H10 (CnH2n+2)

2. BOND-LINE STRUCTURE



2-methylpropane

3. FORMULA CHECK

Counted: C4H10

Rule Check: C4H2(4)+2 = C4H10 ✓

4. BACK-CHECK

✓ **Same as Given**

Returned: 2-methylpropane

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

CARD B02 NAME → STRUCTURE

Foundation | Given: 3-ethylpentane

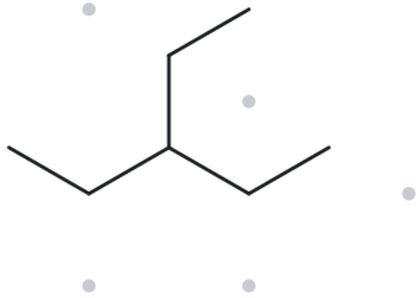
1. NAME DECODED

Parent: pentane (5-carbon chain)

Substituents: ethyl at C3

Expected Formula: C7H16 (CnH2n+2)

2. BOND-LINE STRUCTURE



3-ethylpentane

3. FORMULA CHECK

Counted: C7H16

Rule Check: C7H2(7)+2 = C7H16 ✓

4. BACK-CHECK

✓ Same as Given

Returned: 3-ethylpentane

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

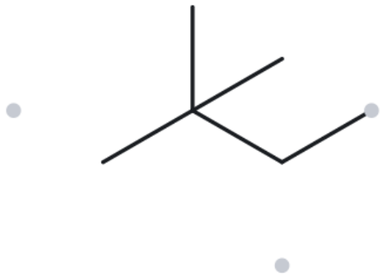
1. NAME DECODED

Parent: butane (4-carbon chain)

Substituents: two methyls at C2

Expected Formula: C6H14 (CnH2n+2)

2. BOND-LINE STRUCTURE



2,2-dimethylbutane

3. FORMULA CHECK

Counted: C6H14

Rule Check: C6H2(6)+2 = C6H14 ✅

4. BACK-CHECK

✓ Same as Given

Returned: 2,2-dimethylbutane

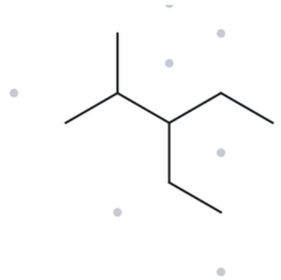
1. NAME DECODED

Parent: pentane (5-carbon chain)

Substituents: methyl at C2, ethyl at C3

Expected Formula: C8H18 (CnH2n+2)

2. BOND-LINE STRUCTURE



3-ethyl-2-methylpentane

3. FORMULA CHECK

Counted: C8H18

Rule Check: C8H2(8)+2 = C8H18 ✓

4. BACK-CHECK

✓ Same as Given

Returned: 3-ethyl-2-methylpentane

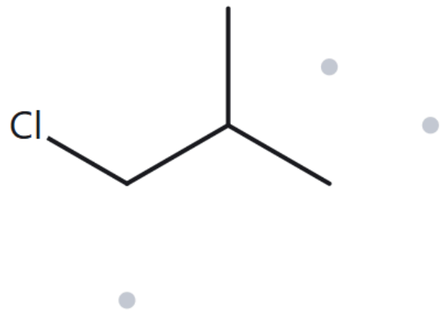
1. NAME DECODED

Parent: propane (3-carbon chain)

Substituents: chloro at C1, methyl at C2

Expected Formula: C4H9Cl (1 H replaced by Cl)

2. BOND-LINE STRUCTURE



1-chloro-2-methylpropane

3. FORMULA CHECK

Counted: C4H9Cl

Rule Check: C4H10 - 1H + 1Cl = C4H9Cl ✅

4. BACK-CHECK

✓ Same as Given

Returned: 1-chloro-2-methylpropane

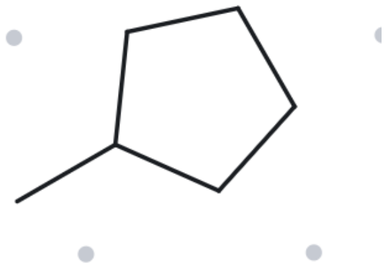
1. NAME DECODED

Parent: cyclopentane (5-carbon ring)

Substituents: methyls at C1, C2

Expected Formula: C7H14 (CnH2n for 1 ring)

2. BOND-LINE STRUCTURE



methylcyclopentane

3. FORMULA CHECK

Counted: C7H14

Rule Check: C7H2(7) = C7H14 ✅

4. BACK-CHECK

✓ Same as Given

Returned: 1,2-dimethylcyclopentane

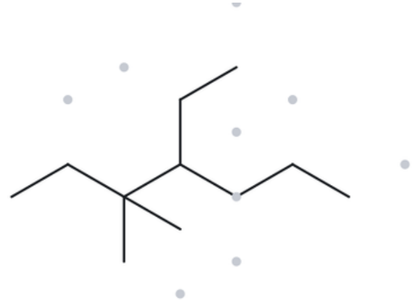
1. NAME DECODED

Parent: heptane (7-carbon chain)

Substituents: methyls at C3, C3; ethyl at C4

Expected Formula: C11H24 (CnH2n+2)

2. BOND-LINE STRUCTURE



4-ethyl-3,3-dimethylheptane

3. FORMULA CHECK

Counted: C11H24

Rule Check: C11H2(11)+2 = C11H24 ✓

4. BACK-CHECK

✓ Same as Given

Returned: 4-ethyl-3,3-dimethylheptane

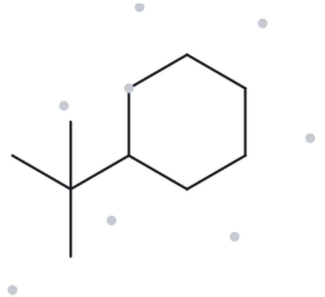
1. NAME DECODED

Parent: cyclohexane (6-carbon ring)

Substituents: tert-butyl group

Expected Formula: C10H20 (CnH2n for 1 ring)

2. BOND-LINE STRUCTURE



(1,1-dimethylethyl)cyclohexane

3. FORMULA CHECK

Counted: C10H20

Rule Check: C10H2(10) = C10H20 ✓

4. BACK-CHECK

✓ **Equivalent Form**

Returned: (1,1-dimethylethyl)cyclohexane

CARD B09 NAME → STRUCTURE

Challenge | Given: 3-ethyl-1,1-dimethylcyclohexane

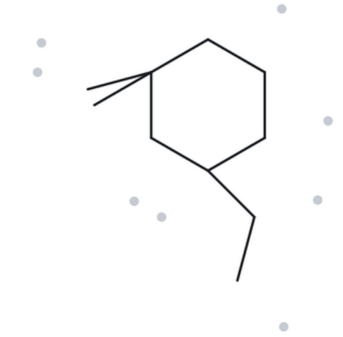
1. NAME DECODED

Parent: cyclohexane (6-carbon ring)

Substituents: methyls at C1, C1; ethyl at C3

Expected Formula: C10H20 (CnH2n for 1 ring)

2. BOND-LINE STRUCTURE



3-ethyl-1,1-dimethylcyclohexane

3. FORMULA CHECK

Counted: C10H20

Rule Check: C10H2(10) = C10H20 ☑

4. BACK-CHECK

✓ Same as Given

Returned: 3-ethyl-1,1-dimethylcyclohexane

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

CARD B10 NAME → STRUCTURE (RULE-TESTING CASE)

Challenge | Given: 2-ethylbutane (Incorrect)

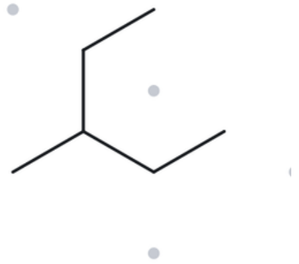
1. NAME DECODED

Given Name: 2-ethylbutane (breaks rules)

Drawn Parent: butane chain with C2 ethyl

Expected Formula: C6H14 (CnH2n+2)

2. BOND-LINE STRUCTURE



3-methylpentane

3. FORMULA CHECK

Counted: C6H14

Rule Check: C6H2(6)+2 = C6H14 ✅

4. BACK-CHECK & RULE ERROR

✕ **Mismatch Discovered**

Returned: 3-methylpentane

Rule Broken: Missed longest chain. Ethyl on C2 extends parent to 5 carbons (pentane).

✓ parent length corrected ✓ substituent positions match ✓ formula agrees ✓ rule error documented